

Fundamentals of AI and ML

(CSA2001)

README : AI Travel Planner

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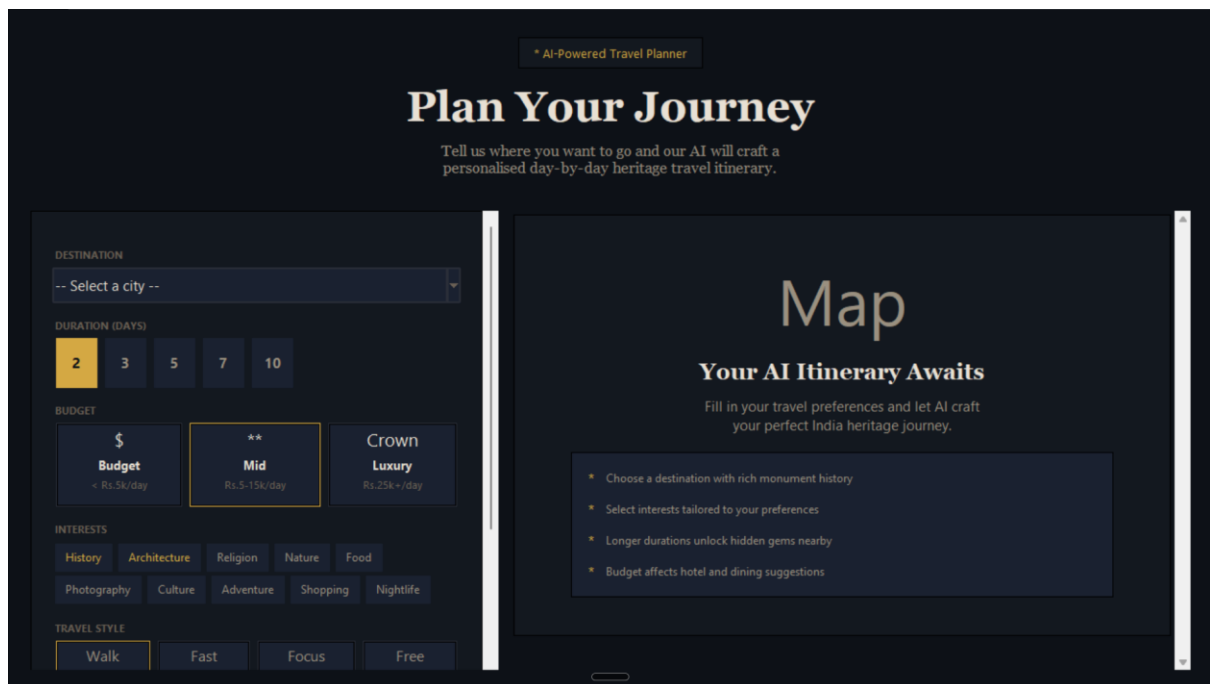
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Branch: Computer Science & Engineering

Specialisation: Artificial Intelligence and Machine Learning

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AI Travel Planner is a premium Python-based desktop application designed to simplify the creation of heritage-focused travel itineraries. Using a custom-themed interface and a robust PDF engine, it transforms travel data into structured, time-blocked plans.



Project Overview

This application caters to travellers seeking cultural depth. It features a curated knowledge base of over 15 major Indian heritage hubs, including **Agra, Delhi, Jaipur, Hampi, Varanasi, etc.**

Key Features

- **Intelligent Scheduling:** Automatically generates detailed itineraries for 2, 3, 5, 7, or 10 days.

- **Time-Blocked Planning:** Daily schedules are divided into **Morning** (7 AM), **Afternoon** (1 PM), and **Evening** (6 PM) slots.
- **Budget Tiering:** Integrated cost estimations for **Budget**, **Mid-range**, and **Luxury** travel styles.
- **Professional PDF Export:** Dynamically creates branded A4 PDF documents using the ReportLab engine.
- **Modern UI:** Features a sleek, dark-themed dashboard with antique gold accents.
- **Simulation Engine:** Provides real-time feedback during processing, such as "Mapping heritage sites by era" and "Optimising travel distances".

The image shows a dark-themed travel planning dashboard. It features several sections with interactive elements:

- DESTINATION:** A dropdown menu with the placeholder text "-- Select a city --".
- DURATION (DAYS):** Five buttons labeled 2, 3, 5, 7, and 10. The button for '2' is highlighted with a yellow border.
- BUDGET:** Three buttons labeled Budget, Mid, and Crown. The 'Mid' button is highlighted with a yellow border. Below each button is a price range: Budget (< Rs.5k/day), Mid (Rs.5-15k/day), and Crown (Rs.25k+/day).
- INTERESTS:** A grid of buttons labeled History, Architecture, Religion, Nature, Food, Photography, Culture, Adventure, Shopping, and Nightlife. The 'Architecture' button is highlighted with a yellow border.
- TRAVEL STYLE:** Four buttons labeled Walk, Fast, Focus, and Free. The 'Walk' button is highlighted with a yellow border.



Technical Stack

- **Language:** Python 3.x
- **GUI Framework:** Tkinter (Custom themed)
- **PDF Generation:** ReportLab
- **Concurrency:** Threading (ensures the UI remains responsive during document generation)
- **Utilities:** Webbrowser, Datetime, OS, Tempfile

Installation & Setup

1. Clone the Repository:

Bash

```
git clone https://github.com/Aklesh1206/Travel-Planner.git
```

```
cd Travel-Planner
```

2. Install Dependencies:

Bash

```
pip install reportlab
```

3. Run the Application:

Bash

```
python Travel.py
```



How to Use

1. **Select Destination:** Choose a city from the dropdown menu (e.g., Delhi, Mysore, or Mumbai).
2. **Set Preferences:** Choose the trip duration and pace (Relaxed, Packed, Focused, or Flexible).
3. **Generate:** Click "Generate Plan" to begin the AI simulation steps.
4. **View/Export:** Once complete, open the professional PDF or share a digital link to view it in a browser.